

Needs Expression / Problem Statement

Port Scan Reconnaissance Detection using Machine Learning

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Network infrastructures are increasingly exposed to **Port Scan Reconnaissance** attacks the systematic first phase of any targeted intrusion, through which attackers map active hosts, open ports, and exposed services before launching a deeper breach. Current intrusion detection systems rely on static, signature-based rules that consistently fail against slow and stealthy scan patterns, while generating a high volume of false positives that overwhelm security teams and delay response to real threats.

Given these limitations, the central problem is formulated as follows:

How can a machine learning model be designed and evaluated to reliably classify network traffic and detect Port Scan Reconnaissance attacks, while drastically minimizing the false positive rate?

Two solution orientations are proposed to address this need: an **automation-oriented** approach, where the model autonomously detects and blocks the suspicious source IP in real time; and a **decision-support-oriented** approach, where the system alerts the network administrator through a dashboard presenting the full attacker profile, leaving the final blocking decision to the human operator.
